Cotyledon

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Executive Summary · Axiom Delta

The sustainable business operating system for microgreens and sprouts farms. Cotyledon runs a farm from the standing order back to the seed: it turns the week's orders into a sow schedule, a daily task list for the crew, a purchasing list and a true cost per tray. It keeps lot traceability and produce safety records, the farm's full financial accounting from each harvest through to the profit and loss statement and balance sheet, and sustainability metrics that include the carbon emissions and water footprint of every tray. A grower adding accounts and territory sees what each new commitment takes in trays, shelf places, labor hours and dollars before making it.

What it answers for a restaurant grower who is scaling

The method lives in a few people's heads

Seeding density, soak, days under weight, blackout, photoperiod, PPFD and watering method by stage are written into each grow plan. A new hire works from the day's task list, and the operations manager stops being the only copy of the method.

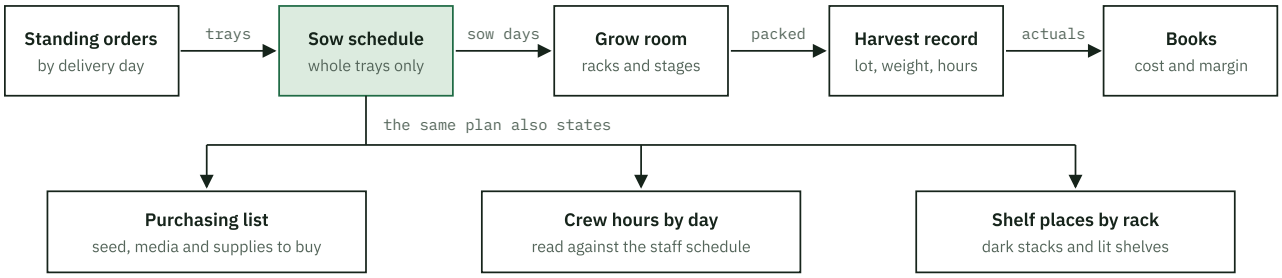
Every new account changes the week

Add a restaurant's standing order and the sow schedule, the rack loading, the crew hours and the seed order all update together. The room's limits show up on screen before they show up as a short delivery.

New territory has to pay for itself

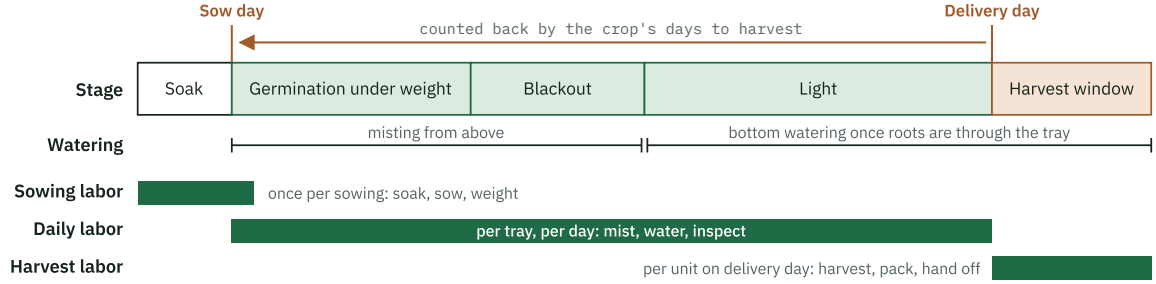
Cotyledon tracks centralized or localized farms, whether one grow room supplies several cities or each city has its own, so you can expand territory efficiently and profitably.

One chain, from order to books



Nothing is entered twice. The orders set the sow schedule, the sow schedule sets the work, the purchasing and the rack loading, and the harvest record closes each lot into the books at what it cost to grow.

Every tray is counted back from its delivery day



Stage lengths are each crop's own; the widths here are schematic. Labor is timed on three streams because they scale differently: sowing with the number of sowings, daily care with trays on the racks, harvest with units delivered.

What the system does, and what the grower gets from it

Grow plans

Each variety or blend carries its seeding density by tray format, medium, nutrient regime, light spec and stage schedule. The same tray, whoever sows it.

Sow schedule from orders

Standing orders are counted back to sow days, in whole trays, net of what is already on the racks. Fewer shorts, less overgrow.

Grow calendar and daily tasks

Every sowing's stages laid across the days: what gets soaked, stacked, uncovered, moved to light and watered today. A crew that knows the day's work.

Capacity by rack and shelf

Dark stacks and lit shelves counted in tray places by format, against what the orders need. Know when the next rack is due.

Labor standards from time studies

Minutes per sowing, per tray per day and per unit, timed on the farm's own crops and averaged as studies are added. Staff to the plan, not to a guess.

True cost per tray

Seed at the last price paid, medium, nutrients, light energy, supplies and labor, over observed yield. Margin by crop, format and account.

Lot traceability and produce safety

One sowing is one lot, from the seed lot received to the tray delivered, with seed sanitation, spent irrigation water tests for sprouts, and temperature and humidity on record. Ready for a group's audit.

Customers, orders and delivery

Standing orders on a weekly, two-week or monthly cadence, delivery routes, tray returns, invoices and card billing, with a portal for customers. Less order-taking by text.

Trials before the menu

A new variety or a chef's custom blend runs as a recorded trial; yield is read across runs before it is offered. Promise only what grows reliably.

CPA-grade accounting

Double-entry books under US GAAP with inventory at actual cost: profit and loss, balance sheet, cash flow, receivables, payables and plan against actual. Books ready to hand to a CPA.

Sustainability metrics

Carbon footprint by scope in CO2e and water consumed per tray, built from energy, inputs, logistics and waste. Numbers a hospitality group can report.

Inventory and purchasing

Seed and media held by lot and used first in, first out, with purchase orders raised from the plan. No sow day short of seed.

Every figure in the system is labeled as measured, quoted by a supplier or estimated.

Portals for customers and sales

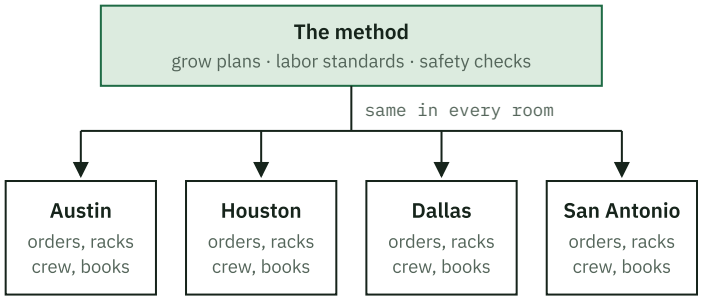
Customer Portal

Each customer signs in to their own account to see standing orders, upcoming deliveries and invoices, keep a card on file, and skip or pause a delivery up to its sow date. A change to a standing order arrives as a request the farm approves. Customers serve themselves; the farm keeps control of the plan.

Sales CRM Portal

The sales team works each prospect from first contact to quote and scope of work, with the pipeline, the customers on file and the upcoming orders on one screen. A prospect who signs becomes a customer whose standing order feeds the sow schedule. Sales and production read the same numbers.

One method, central or local farms



A local farm in each city is shown here; one central farm supplying several cities is tracked the same way. Each farm has its own orders, racks, crew and books.

Where it comes from

Robert Bogatin built Cotyledon from his own production records at Vallecito Micro Farm in Bayfield, Colorado, where he grew twelve varieties and specialized in trays composed for individual clients. The system holds his method: each growing and nutrition claim in it cites a registered source, and each standard is measured on the farm's own crops. It runs on Ember OS, the platform behind Muse Kitchen, his operating system for commissary kitchens.

Where it stands

Cotyledon is in active development: the planning, production and costing core runs today, and a few pieces, such as delivery routes and staffing, are still being finished.

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